

GazeClass: Towards Gaze-Adaptive Cross-Device Learning Support for Virtual Classrooms

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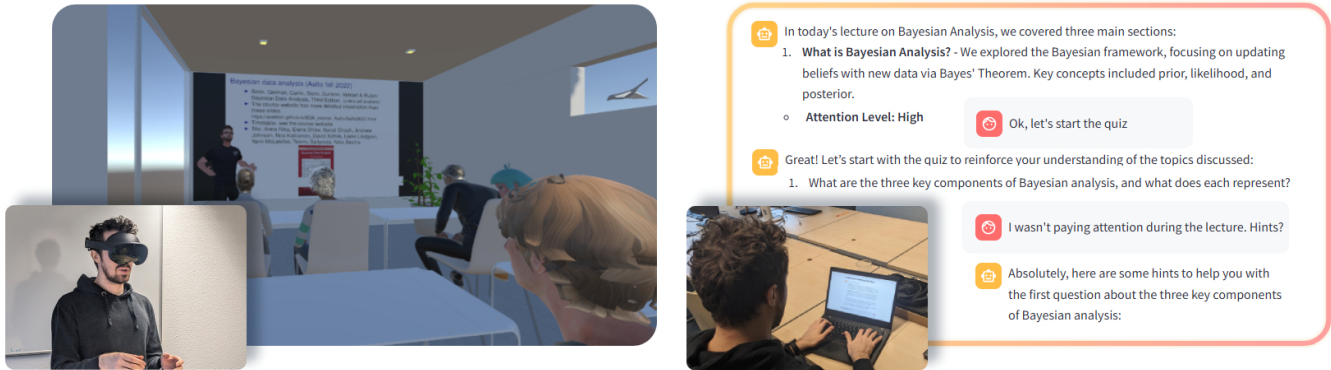


Figure 1: GazeClass presents a VR lecture (left), where eye tracking data is collected to assess users' attention levels during the session, and a web-based assistant (right) that delivers a personalized quiz after the lecture, tailored to users' attention patterns.

Abstract

Immersive virtual classrooms provide engaging learning experiences but often lack individualized support, particularly in asynchronous settings where students independently engage with pre-recorded lectures. This absence of live teacher interaction can hinder personalized guidance and self-reflection. To address this challenge, we present *GazeClass*, a system enabling cross-device interaction by combining eye tracking in Virtual Reality (VR) with a learning assistant powered by large language models (LLMs) on a separate mobile or desktop device, delivering post-lecture, gaze-adaptive learning support. During the lecture, eye tracking monitors students' attention, and afterward, tailored feedback and self-reflective Q&A are provided to bridge attention gaps. The cross-device interaction between VR and a mobile or desktop device fosters interactivity and active learning. Developed through co-design workshops and iterative evaluations, *GazeClass* was validated in a preliminary study, demonstrating its potential to enhance learning outcomes by addressing key challenges in immersive learning.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality.**

Keywords

mixed reality, eye tracking, virtual classroom, adaptive support

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1 Introduction and Related Work

Immersive virtual learning environments have been designed and evaluated across diverse educational contexts [21, 29, 32]. In particular, immersive lectures conducted within VR-based virtual classrooms exemplify the potential to enhance students' engagement [11, 22]. Virtual classrooms enable students to consume lectures in an immersive environment, either collaboratively or individually [11]. Additionally, virtual classrooms provide instructors with tools to demonstrate concepts in tangible and dynamic ways [19]. Collectively, existing research underscores the potential of virtual classrooms to foster active learning and improve outcomes [10, 11, 18].

Nonetheless, immersive learning also has limitations and poses challenges for users. Support plays a crucial role in the learning

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process [30]. In traditional, non-immersive lectures, support is commonly facilitated through teacher-student interaction [1]. In VR-based virtual classrooms, synchronous learning sessions can replicate these interactions, enabling real-time teacher-student engagement [17, 33]. However, many immersive learning environments are designed for asynchronous use, where students engage with pre-recorded content independently [24]. In such settings, the absence of live teacher interaction can make it challenging for students to receive individualized support. To address this, post-lecture support (e.g., tutoring sessions with Q&A) has been proposed as a viable solution for providing learning support in asynchronous scenarios [2].

Post-lecture support is recognized as an effective method for improving learning outcomes [34]. Beyond VR-based virtual classrooms, both human tutoring and computer-based intelligent tutoring systems (ITS) have demonstrated effectiveness in enhancing students' comprehension of learning content [7, 36]. Recent studies have also explored the implementation of learning support in virtual classrooms [2, 33]. However, the effectiveness of these systems remains constrained by their limited capacity to deliver highly personalized support. For achieving more personalized experiences, gaze-adaptive systems, mainly available with desktop eye tracking, have showcased promising support mechanism such as attention management during learning [8, 20]. Nonetheless, in immersive learning, the application of eye tracking has been predominantly limited to gaze-based interaction and usability evaluation [28], with little exploration into gaze-adaptive support.

To address these gaps, this paper presents *GazeClass*, a cross-device learning system that enhances VR-based lectures. In *GazeClass*, an embodied lecturer avatar delivers a pre-recorded lecture within a virtual classroom. Eye tracking data collected with the VR headset is seamlessly integrated with a web-based learning assistant accessible via mobile or desktop devices. This assistant allows learners to receive personalized feedback and engage in a tailored Q&A session driven by an LLM after the lecture. We iteratively developed and evaluated *GazeClass* through a series of studies. This included a co-design workshop ($n = 6$) to identify user challenges in immersive learning, followed by an exploratory evaluation in the wild ($n = 7$) conducted as part of university lectures. After a design iteration, we conducted a preliminary study ($n = 9$) to evaluate the updated system design. Our findings suggest that: (1) post-lecture support in virtual classrooms can enhance learning experiences, aligning with prior research; and (2) gaze-adaptive learning support, as implemented in *GazeClass*, shows promise for improving learning outcomes.

We aim to contribute to the HCI community in the following ways: First, we designed, implemented, and evaluated a novel system that integrates gaze-adaptive learning support in a virtual classroom context following a cross-device approach. Second, we offer design implications and insights on human-centered learning support in virtual classrooms. Third, we address the gap in using eye tracking data as attentional input to inform LLM-based learning support.

Immersive Learning Systems and Virtual Classrooms Immersive learning creates interactive learning experiences with a higher level of immersion and sense of presence, utilizing technologies such as VR [27]. Among immersive learning environments,

VR-based virtual classrooms present a unique learning experience by replicating aspects of traditional classroom learning, including peer interaction [22] and teacher-student engagement [39]. Despite their advantages, user challenges persist, and HCI research has sought to address these challenges. For example, Han et al. [13] explore intervention mechanisms to mitigate student distraction using hints and warnings. However, learners' support in virtual classrooms, especially personalized support, remains an underexplored area [23, 39].

Eye tracking and Gaze-adaptive Learning Support As a complex cognitive process, learning can benefit from adaptive systems that tailor support to individual needs [16]. In specific, gaze-adaptive systems provide cognitive assistance by offering attention feedback with attentive user interfaces [5, 9]. While eye tracking is increasingly accessible on VR devices [38], using it for gaze-adaptive learning support requires further research.

2 Co-Design Workshop

We investigate the following research question: *How to design effective learning support for learners in VR-based virtual classrooms?* To answer this, we employed a human-centered approach, beginning with a co-design workshop involving six industrial engineering students (average age: 23 years, $SD = 1.63$, 2 females, 4 males), with two participants having prior experience with immersive technologies.

2.1 Task and Procedure

This co-design workshop aimed to identify existing and potential user challenges in immersive learning systems and formulate design goals for effective support interventions in immersive learning experiences, particularly within virtual classrooms. Six participants were randomly divided into two groups and engaged in the following activities (See Appendix A.1 for the complete protocol):

Onboarding. An onboarding session introduced participants to basic interaction techniques across various immersive technologies, such as gesture- and gaze-based interactions with MR and VR headsets (e.g., HoloLens and Meta Quest Pro). Subsequently, participants were introduced to three existing immersive learning artifacts. These artifacts, encompassing diverse immersive technologies (MR, VR, mobile AR), designs, and learning content (human information processing, neural networks, K-means clustering), aimed to minimize the priming effect of any single technology or design on participants' impressions of immersive learning. The onboarding session lasted 10 minutes for each group.

Think-Aloud. Following the onboarding, each participant was assigned a learning task with one of the immersive learning artifacts (e.g., exploring all components of the human information processing model using gaze interaction). The tasks were designed to take approximately 15 minutes to complete. Participants were instructed to employ the think-aloud technique, verbalizing their thoughts and challenges throughout the task. These think-aloud sessions were documented in protocols according to Charters [6]. Once a participant completed their assigned task, they proceeded to the next artifact. The think-aloud sessions concluded when all participants had completed tasks with all three artifacts. The sessions lasted 45 and 51 minutes for the two groups.

Focus Group Discussion. Following the think-aloud sessions, participants engaged in a focus group discussion guided by the framework proposed by Mazza and Berre [25]. The discussion focused on participants reflecting on their think-aloud protocols and elaborating on their challenges and mental models when interacting with the immersive learning artifacts. Furthermore, participants shared learning support strategies employed during traditional lectures and discussed their potential adaptability to immersive learning environments. Two focus groups participated in discussions that lasted 11 minutes and 15 minutes, respectively.

Brainwriting. After the discussion, participants engaged in a co-design session using the 6-3-5 brainwriting method. This ideation technique, originally proposed for product innovation [15, 35], has been widely adopted in interdisciplinary and HCI research as an effective participatory design method [3, 26].

In our adapted version, each participant was given a worksheet and 10 minutes to write or draw ideas before passing their worksheet to the next participant. This process was repeated for three rounds until each participant received their initial worksheet back, having reviewed ideas from all other participants.

2.2 Results



Figure 2: (a) Participants tested existing applications while engaging in think-aloud; (b) the focus-group discussion and brainwriting session; (c) - (d) design ideas produced during brainwriting

We analyzed two key data sources from the design workshop. First, the think-aloud protocols were reviewed alongside the transcriptions of the focus group discussion. Two coders independently coded the data and concluded three user challenges in a thematic analysis following the methodology proposed by Braun and Clarke [4]. The analysis provided insights into the existing challenges during immersive learning experiences. Second, the worksheets completed by participants during the 6-3-5 brainwriting session were analyzed to identify potential design solutions. These insights were then used to formulate concrete design goals for effective learning support in virtual classrooms.

2.2.1 User Challenges. The thematic analysis revealed three key user challenges as themes:

First, immersive learning experiences often lack **interactive support**. Most Participants (5 out of 6) highlighted the difficulty of accessing interactive support during VR-based lectures, unlike in Massive Open Online Courses (MOOCs) where they could easily search for information or use tools like ChatGPT. For instance, one participant mentioned that they “would Google something or use ChatGPT after the lecture” (P2). However, in a VR learning environment, obtaining such interactive support is not as straightforward as “opening another tab in the browser window” (P5). Overall, participants expressed the need for a seamless and non-intrusive way to access interactive support for immersive learning.

While providing real-time support during the lecture was considered, **managing mental workload** emerged as a second significant challenge. Participants reported that immersive learning experiences inherently require mental effort for active engagement. Therefore, introducing additional real-time support during the lecture in VR could be disruptive. Many participants (4 out of 6) believed that using novel technologies like VR can already pose difficulties for first-time users unfamiliar with the interaction modalities. They expressed concerns that intrusive support features, such as pop-up notifications during an immersive lecture, might create cognitive overload and hinder the learning process.

Lastly, participants pointed out a **lack of personalization** in immersive learning. While all participants appreciated the high interactivity and immersion of the experience, three participants (P2, P3, P5) noted the absence of tailored assistance based on individual learning progress. They compared the experience unfavorably to traditional learning methods, such as tutoring or taking notes, which offer “tailored and personal” (P3) self-reflection after lectures.

2.2.2 Design Goals. we formulate the following design goals after the workshop:

Prioritize Post-Lecture Support: To mitigate the potential cognitive load associated with adding real-time support during the lecture, the primary design goal is to prioritize post-lecture support mechanisms. This approach allows learners to fully engage with the VR experience without distraction and provides a more conducive environment for information processing and reflection. Post-lecture support could include personalized summaries based on user interactions within the VR environment and tailored questions in Q&A sessions.

Integrate Cross-Device Interaction: To address the limitations of accessing support within VR, the system should integrate cross-device interaction. This allows learners to seamlessly transition from the immersive VR experience to a companion application on a mobile device or desktop computer, enabling easier access. During the workshop, five participants highlighted the effectiveness of chat-based assistance, such as ChatGPT, and suggested that combining immersive learning with web-based tools could offer intuitive support without overwhelming first-time users.

Explore Gaze-Adaptive Support: Two participants (P3, P6) proposed utilizing eye tracking for learning analytics. Since our primary focus is on providing post-lecture support, we aim to synchronize eye tracking data collected during the lecture with a web-based assistant. During the lecture, the system should analyze learners’ gaze data to infer attention levels. This data will then be used to inform and personalize the web-based assistant afterward, enhancing self-reflection and comprehension for learners.

3 GazeClass: Design and Exploratory Evaluation

3.1 Initial Design

GazeClass integrates two primary components: a VR lecture scene implemented using Unity 3D (2022.3.30f1) for VR headset, such as Meta Quest Pro; and a web-based learning assistant, accessible via desktop or mobile devices.

The VR lecture scene simulates a standard virtual classroom environment featuring an embodied avatar delivering a prerecorded

lecture. The audio track is synchronized with the avatar's lip movements using Oculus Lipsync¹, while lecture slides are displayed and synchronized with the lecture content (See Fig. 3 - Initial Design). The lecture content is divided into multiple sections based on the topics to facilitate later processing. During the lecture, the system leverages eye tracking data from the VR headset to measure attention levels. When users focus their gaze on the avatar or lecture slides, the system logs focused learning time. If the user's gaze moves away from these areas of interest, the tracking pauses. At the end of the lecture, the accumulated focused time for each section is recorded in a CSV file and transmitted to a Flask server via HTTP for subsequent use.

The web-based learning assistant is implemented with Streamlit², which reads the CSV file and leverages the attention data as well as the lecture transcript to provide personalized post-lecture support. Integrated with the OpenAI API (GPT-4o-mini model), the assistant presents a tailored summary of the lecture, highlighting the user's attention distribution across all sections. It then generates quiz questions, prioritizing sections with lower attention levels while ensuring comprehensive coverage by including at least one question per section. Users interact with the assistant to answer questions, receive hints, and access detailed explanations (See Fig. 1).

3.2 Exploratory Evaluation in the Wild

3.2.1 Method. We conducted an in-the-wild study at our local university in collaboration with two courses: one on business intelligence systems and the other on biosignal-adaptive systems. Both courses included self-study sessions requiring students to independently watch pre-recorded lectures. With the lecturers' coordination, we invited students from both courses to voluntarily participate in a session using *GazeClass* to complete one self-study lecture. Seven participants (average age: 23.1 years, SD = 1.57, 2 females, 5 males) were recruited, including four industrial engineering majors and three information systems majors. Two participants reported prior experience with mixed reality devices.

Participants were provided with a Meta Quest Pro headset to view the lecture in VR. The pre-recorded lectures and slides, supplied by the lecturers, were integrated into the VR lecture scene. To mitigate potential motion sickness and cognitive overload, the lectures were trimmed to an average duration of 15 minutes. Each lecture, confirmed by the lecturers, was transcribed and divided into three sections based on learning topics as the knowledge base of the learning assistant. After consuming the lecture in VR, participants removed the headset and could choose between an Android smartphone or a Windows PC to interact with the learning assistant. As described in the previous subsection, the assistant provides students with a personalized summary and quiz questions, prioritizing sections with lower attention levels.

For each session, user data was collected with informed consent, including total focused time during the VR lecture, usage time of the learning assistant, and qualitative feedback gathered through a 10-minute post-session interview.

3.2.2 Results. All seven participants successfully completed the self-study session using *GazeClass*. Their log data revealed a mean focus rate during the lecture in VR, calculated as the total focused time divided by the total lecture length, of 86.1% (SD = 7.9%) and an average interaction time with the learning assistant of 280 seconds (SD = 126.08).

A thematic analysis of the interview transcripts highlighted two main findings. First, participants noted the **effectiveness of personalization**. Four participants (P2, P4, P5, P6) reported that the personalized questions were highly relevant and accurately addressed their areas of low attention. The other participants, with a focus rate exceeding 95%, expressed less need for post-lecture Q&A. Second, all participants appreciated the system's ability to **guide self-reflection**. They compared the experience to attending a tutoring session, emphasizing how it allowed them to review at their own pace while still feeling "guided in the process" (P5).

Despite these positive impressions, participants suggested improvements to enhance the accuracy of the attention-tracking algorithm and recommended testing the system with different lecture content to assess its scalability.

4 Preliminary Study

4.1 Iterative Redesign of GazeClass

Following the exploratory evaluation, several refinements were made to *GazeClass* to address participants' feedback. One change was the replacement of the original lecture content with a 48-minute introductory lecture on Bayesian analysis by Vehtari [37], a widely recognized resource integrated into different university curriculums [12]. This adjustment ensured that the lecture length and content better represented real-world scenarios and reduced potential bias stemming from participants' prior knowledge of the material. In addition, to foster a stronger sense of social presence, we added six animated peer avatars seated across two rows of tables in the virtual classroom. These non-interactive avatars simulate the collaborative dynamics of a real classroom, addressing feedback that the environment previously lacked social realism.

Additionally, we refined the attention-tracking system to account for "off-slide" segments, where lecturers (including in the Bayesian analysis lecture) provide critical information not displayed on the slides. These segments were manually coded, and attention criteria for these segments were adjusted to only classify students as attentive when their gaze focused on the embodied lecturer avatar rather than the slides (See Fig. 3 for an overview of design iterations).

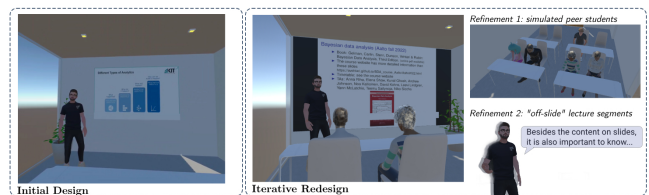


Figure 3: Design comparison between the initial design and the iteration following the exploratory evaluation

¹Oculus Lipsync: <https://developers.meta.com/horizon/documentation/unity/audio-ovrlipsync-unity/>

²<https://streamlit.io/>

4.2 Task and Procedure

We conducted a preliminary study to evaluate the iterative design, approved by our Institutional Review Board (IRB). Each participant attended an individual session and was informed about our data collection and the gaze-adaptive design. After providing informed consent, participants underwent a 15-minute onboarding to familiarize themselves with the virtual classroom environment of *GazeClass*.

Following onboarding, participants completed a pre-survey with a question on their prior knowledge of Bayesian analysis and a NASA-TLX questionnaire [14] to measure mental workload. Participants then attended the 48-minute introductory lecture on Bayesian analysis in VR with a Meta Quest Pro headset. After the lecture, they removed the headset and completed a second NASA-TLX questionnaire to assess their mental workload post-lecture, before interacting with the gaze-adaptive learning assistant on a PC or smartphone.

After completing the quiz generated by the learning assistant, they completed the post-study survey with the following: (1) a knowledge test with 10 questions based on the course assignment of Vehtari [37]; (2) the User Experience Questionnaire (UEQ) [31] to assess the overall user experience; (3) a final NASA-TLX to measure mental workload after using the learning assistant; and (4) a subjective evaluation with a question: *Q1: "How well do you think the generated questions reflected your attention distribution throughout the lecture?"*. The session concluded with a semi-structured interview covering user experience, learning support efficacy, gaze-adaptive features, and potential improvements.

4.3 Results

We recruited nine participants (average age: 24 years, SD = 2.83, 3 females, 6 males). Two participants had prior experience with VR headsets, while the remaining seven were first-time VR users. None of the participants reported prior knowledge of Bayesian analysis. All participants completed the given task, spending an average of 8.21 minutes (SD = 4.17) interacting with the learning assistant after attending the 48-minute lecture.

4.3.1 Quantitative Analysis. During the immersive lecture, participants demonstrated an average focus rate of 74.14% (SD = 0.15). Attention management difficulty was noticeable during the "off-slide" segments, which resulted in a lower average focus rate of 66.64% (SD = 0.18). The perceived accuracy of the post-lecture questions (See Appendix A.5 for the response distribution of Q1) was evaluated on a 5-point Likert scale (1 = inaccurate, 5 = very accurate), with participants giving an average rating of 4.2 (SD = 0.83).

User experience was assessed using the UEQ toolkit by Schrepp et al. [31]. *GazeClass* achieved an average attractiveness score of 2.43 (SD = 0.13), along with high ratings for pragmatic quality (2.08) and hedonic quality (2.32), suggesting an overall positive user experience with the learning assistant (See Appendix A.7).

We conducted a repeated measures ANOVA on NASA-TLX results collected across three points in time: after onboarding, after the lecture, and after using the learning assistant. Results indicated a change in mental workload over time, $F(2, 16) = 45.06, p < .001$,

partial $\eta^2 = 0.849$. Post-hoc pairwise comparisons with Holm's correction showed a significantly higher mental workload after the lecture ($M = 57.78, SD = 9.39$) compared to before the lecture ($M = 15.56, SD = 9.17$), with a mean difference of 42.22 ($p < .001$). While the mental workload after using the assistant ($M = 49.44, SD = 9.17$) was slightly lower than after the lecture, the reduction was not statistically significant ($p = 0.096$). However, significant reductions were observed for the temporal workload ($M_{\text{diff}} = 15, p = 0.003$, Cohen's $d = 1.87$) and physical workload ($M_{\text{diff}} = 15, p < .001$, Cohen's $d = 2.65$), suggesting that the learning assistant helped alleviate some difficulties (See Appendix A.6). The knowledge test revealed an average score of 6.78 (SD = 0.97) out of 10. Given that none of the participants had reported prior knowledge, this indicates that *GazeClass* effectively facilitated learning.

4.3.2 Qualitative Analysis. A thematic analysis of the interviews yielded the following themes:

Avoiding Cognitive Overload with Post-Lecture Support.

Participants emphasized the challenges of maintaining focus during the lecture and found it "difficult to process new information towards the end" (P6). This aligns with the NASA-TLX data, which showed an increased workload after the lecture. Reported as an effective method of supporting them, participants appreciated the post-lecture assistant, with six participants noting that the assistant helped them "really learn something in such a short time" (P7) with tailored quiz questions.

Promoting Self-Reflection with Gaze-Adaptive Assistance.

Participants found the gaze-adaptive features accurate and aligned with their attention allocation during the lecture. The personalized feedback encouraged self-reflection, with three participants (P2, P3, P5) reporting that they actively considered why their attention was low during certain sections. The generated quiz questions were described as "helpful in bridging gaps" in missed content (P5), reinforcing the system's effectiveness.

A Friendly User Experience with Cross-Device Interaction.

Participants appreciated the ease of transitioning from the VR lecture to the web-based learning assistant. They noted that receiving support on familiar platforms (such as with PC or smartphones) enhanced usability and reduced the uncertainty during learning. The process of removing the VR headset and switching to a new device also helped shift their attitude from passive content consumption to a "more active learning mindset" (P1).

5 Limitations and Future Work

Our findings highlight the feasibility of gaze-adaptive learning support delivered after immersive lectures, but several limitations warrant discussion. First, while we focused on post-lecture support to mitigate mental workload during immersive lectures, real-time support during lectures could further enhance learning outcomes. Second, despite improvements in our attention-tracking mechanism, its granularity remains limited. Refining this mechanism could enhance system efficacy. Finally, certain biases may have influenced our findings: the preference of a web-based chat interface reflects participants' potential legacy bias during the design workshop, favoring familiar technologies.

Future work will address some limitations with a between-subjects experiment with a larger sample size to provide a robust assessment

of the system. A baseline comparison with a non-adaptive design will clarify the added value of gaze-adaptive support. Finally, exploring alternative designs, such as integrating an embodied agent within the VR environment as a learning assistant, may uncover new opportunities to enhance engagement and learning outcomes. Lastly, potential privacy risks with eye tracking technology must be considered for future gaze-based learning analytics.

6 Conclusion

In this paper, we presented *GazeClass*, a gaze-adaptive system that provides post-lecture learning support through cross-device interaction. Our findings demonstrate its potential to enhance learning outcomes by delivering personalized summaries and quiz questions tailored to students' attention patterns. While limitations and opportunities for further improvements remain, this work highlights the value of gaze-adaptive technologies in supporting learners and lays a foundation for advancing personalized learning support in immersive learning systems.

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A APPENDICES

A.1 Instructions and protocol for the co-design workshop

Greetings. Welcome to the design workshop on intelligent learning support for immersive learning. This workshop will provide you with hands-on experience with three different XR learning applications:

- Mobile AR: An application teaching K-means clustering.
- Head-mounted MR: A HoloLens application covering human information processing concepts.
- VR: A VR application teaching the basics of neural networks

Think-Aloud Sessions. During your interactions with each application, please engage in “think-aloud” process. This involves verbalizing your thoughts, challenges, and ideas as you use the applications. Share your impressions freely; there are no right or wrong answers.

Tasks for learning applications.

- Mobile AR (K-means Clustering):
 - (1) Start the learning process by scanning the QR code.
 - (2) Explore the visualized data point distribution.
 - (3) Follow the instructions and observe how the centroids update.
 - (4) Repeat the process until the clustering converges.
 - (5) Summarize the key steps of the K-means clustering algorithm.
- Head-mounted MR (Human Information Processing):
 - (1) Follow the introduction scenes to learn about HIP concepts.
 - (2) Enter the explore scene and use gaze-based interaction to trigger learning components.
 - (3) Explore all four components of information processing (motor, cognition, auditory perception, visual perception).
 - (4) Complete the application with attention feedback.
- VR (Neural Network):
 - (1) Familiarize yourself with the VR environment.
 - (2) Open a pre-recorded lecture video using the controller or hand gestures.
 - (3) Follow the lecture instructions to complete a simple neural network visualization
 - (4) Watch the lecture at your own pace and follow the instructions
 - (5) Attempt to memorize important content throughout the lecture.

Focus Group Discussion. After using all three applications, we will engage in a group discussion. Please reflect on:

- (1) Overall Impressions:
 - Challenges encountered while using the applications.
 - Differences and preferences between the XR technologies.
 - Suitability of each learning topic and content for the respective technology.
- (2) Learning Experience:
 - Compare learning with XR to traditional or online methods (MOOCs).

- Identify features that would enhance learning with these XR applications.
- Describe your existing learning support methods (e.g., tutors, exercises, online tools).

Co-Design Task. Imagine you are designing an intelligent tutor for a VR virtual classroom. In groups, brainstorm and sketch ideas on a provided worksheet.

- Use different colors to distinguish ideas from each group member.
- Feel free to pass the worksheet around during the ideation process.
- You have 10 minutes for the co-design task.

Wrap-up Discussion. Briefly present your group's design ideas, explaining the rationale behind each.

A.2 Participant information of the exploratory evaluation

Table 1: Participant information of the exploratory evaluation

Number	Age	Major	Gender	Prior Experience with XR	Education Level
1	24	Information Systems	female	none	Bachelor's Student
2	25	Information Systems	male	none	Bachelor's Student
3	22	Industrial Engineering	female	HoloLens	Bachelor's Student
4	25	Industrial Engineering	male	none	Bachelor's Student
5	21	Information Systems	male	none	Bachelor's Student
6	23	Industrial Engineering	male	none	Bachelor's Student
7	22	Industrial Engineering	male	VR with Meta Quest 3	Master's Student

A.3 Participants information of the preliminary study

Table 2: Participants information of the preliminary study

Number	Age	Major	Gender	Prior Experience with XR	Education Level
1	24	Information Systems	female	none	Bachelor's Student
2	25	Industrial Engineering	female	none	Master's Student
3	23	Information Systems	male	none	Bachelor's Student
4	24	Industrial Engineering	male	none	Master's Student
5	21	Information Systems	male	none	Bachelor's Student
6	26	Mechanical Engineering	male	none	Master's Student
7	22	Industrial Engineering	male	VR with Meta 3	Master's Student
8	27	Product Design	female	VR in museum	Master's Student
9	31	Mechatronics	male	none	PhD Student

A.4 Knowledge test questions in the preliminary study

- (1) What is a 'full probability model' in Bayesian data analysis?
 - A) A model that accounts for all possible outcomes and their probabilities.
 - B) A model that includes both the prior distribution and the likelihood function.
 - C) A model that only considers the likelihood function.
 - D) A model that excludes prior information.
- (2) In Bayesian terminology, what does the symbol θ typically represent?
 - A) Observed data.
 - B) Prior distribution.
 - C) Model parameters.
 - D) Posterior distribution.
- (3) What is the purpose of the prior distribution in Bayesian analysis?
 - A) To represent the data observed.
 - B) To express initial beliefs about the parameters before observing data.
 - C) To calculate the likelihood of the data.
 - D) To normalize the posterior distribution.
- (4) Which of the following best describes the 'posterior distribution'?
 - A) The distribution of the data given the parameters.
 - B) The initial belief about the parameters before observing data.
 - C) The updated belief about the parameters after observing data.
 - D) The distribution of future observations.
- (5) What does the notation $p(y|\theta)$ represent in Bayesian statistics?
 - A) The prior distribution of θ .
 - B) The likelihood of observing data y given parameters θ .
 - C) The posterior distribution of θ given data y .
 - D) The marginal distribution of y .
- (6) In the context of Bayesian inference, what does 'exchangeability' refer to?
 - A) The property that the order of data points does not affect their joint probability.
 - B) The independence of data points.
 - C) The equality of prior and posterior distributions.
 - D) The ability to swap parameters without changing the model.
- (7) What is the role of the likelihood function in Bayesian analysis?
 - A) To provide the prior distribution of the parameters.
 - B) To update the prior distribution with observed data.
 - C) To normalize the posterior distribution.
 - D) To predict future data points.
- (8) Which of the following symbols is commonly used to denote 'distributed as' in statistical notation?
 - A) $\propto$
 - B) $\sim$
 - C) θ
 - D) $p(\cdot|\cdot)$
- (9) What does the symbol $\propto$ signify in Bayesian equations?
 - A) Equality between two expressions.
 - B) Proportionality between two expressions, indicating they are equal up to a constant factor.
 - C) Independence between variables.
 - D) Distribution of a random variable.
- (10) In Bayesian analysis, what is the 'posterior predictive distribution'?
 - A) The distribution of the parameters after observing data.
 - B) The distribution of future observations based on the current model.
 - C) The prior distribution updated with new data.
 - D) The likelihood function normalized over all possible data.

A.5 Knowledge test and response to Q1 in the preliminary study

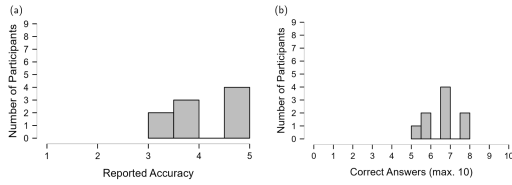


Figure 4: (a) Distribution of participants' responses to Q1 on their perceived accuracy of the gaze-adaptive quiz generation (b) Distribution of participants' scores in the knowledge test in the post-survey

A.6 Repeated measures ANOVA of NASA-TLX in the preliminary study

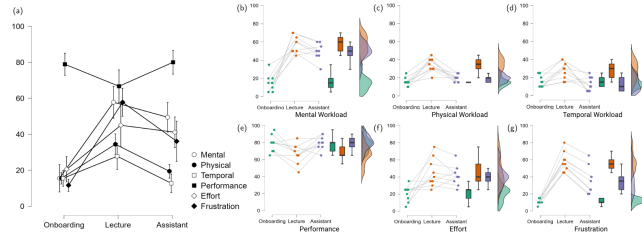


Figure 5: (a) Overview of the trends in the six NASA-TLX dimensions (Mental Workload, Physical Workload, Temporal Workload, Performance, Effort, and Frustration) across three points in time during the preliminary study: After onboarding, after lecture, and after using the learning assistant. (b–g) Raincloud plots for each NASA-TLX dimension, illustrating individual participant responses (lines), the overall distribution, and central tendency of the reported values at each timepoint.

Table 3: Within subjects effects - mental workload over time

Cases	Sum of Squares	df	Mean Square	F	p
Time	9001.852	2	4500.926	45.061	< .001
Residuals	1598.148	16	99.884		

Table 4: Post hoc comparisons - mental workload over time

		Mean Difference	SE	t	<i>P</i> _{Holm}
Onboarding	Lecture	−42.222	4.711	−8.962	< .001
	Assistant	−33.889	4.711	−7.193	< .001
Lecture	Assistant	8.333	4.711	1.769	0.096

Table 5: Within subjects effects - physical workload over time

Cases	Sum of Squares	df	Mean Square	F	p
Time	1790.741	2	895.370	36.491	< .001
Residuals	392.593	16	24.537		

Table 6: Post hoc comparisons - physical workload over time

		Mean Difference	SE	t	Cohen's d
Onboarding	Lecture	−18.889	2.335	−8.089	−3.342
	Assistant	−3.889	2.335	−1.665	−0.688
Lecture	Assistant	15.000	2.335	6.424	2.654

Table 7: Within subjects effects - temporal workload over time

Cases	Sum of Squares	df	Mean Square	F	p
Time	1146.296	2	573.148	9.450	0.002
Residuals	970.370	16	60.648		

Table 8: Post hoc comparisons - temporal workload over time

		Mean Difference	SE	t	Cohen's d
Onboarding	Lecture	−12.222	3.671	−3.329	−1.529
	Assistant	2.778	3.671	0.757	0.348
Lecture	Assistant	15.000	3.671	4.086	1.877

Table 9: Within subjects effects - performance over time

Cases	Sum of Squares	df	Mean Square	F	p
Time	985.185	2	492.593	5.919	0.012
Residuals	1331.481	16	83.218		

Table 10: Post hoc comparisons - performance over time

		Mean Difference	SE	t	<i>P</i> _{Holm}
Onboarding	Lecture	12.222	4.300	2.842	0.024
	Assistant	−1.111	4.300	−0.258	0.799
Lecture	Assistant	−13.333	4.300	−3.101	0.021

Table 11: Within subjects effects - effort over time

Cases	Sum of Squares	df	Mean Square	F	p
Time	3105.556	2	1552.778	9.894	0.002
Residuals	2511.111	16	156.944		

Table 12: Post hoc comparisons - effort over time

		Mean Difference	SE	t	<i>p</i> _{holm}
Onboarding	Lecture	−24.444	5.906	−4.139	0.002
	Assistant	−20.556	5.906	−3.481	0.006
Lecture	Assistant	3.889	5.906	0.659	0.520

Table 13: Within subjects effects - frustration over time

Cases	Sum of Squares	df	Mean Square	F	p
Time	3105.556	2	1552.778	9.894	0.002
Residuals	2511.111	16	156.944		

Table 14: Post hoc comparisons - frustration over time

		Mean Difference	SE	t	<i>p</i> _{holm}
Onboarding	Lecture	−46.111	3.826	−12.053	< .001
	Assistant	−24.444	3.826	−6.390	< .001
Lecture	Assistant	21.667	3.826	5.664	< .001

A.7 UEQ Result in the preliminary study

Table 15: Overview of all UEQ items with mean, variance, standard deviation, and their corresponding scales

Mean	Variance	Std. Dev.	Descriptive Pairs	Corresponding Scale
2.0	1.3	1.1	annoying – enjoyable	Attractiveness
2.4	0.8	0.9	not understandable – under.	Perspicuity
2.3	0.5	0.7	creative – dull	Novelty
2.3	0.8	0.9	easy to learn – difficult to learn	Perspicuity
2.0	2.5	1.6	valuable – inferior	Stimulation
2.6	0.5	0.7	boring – exciting	Stimulation
2.2	1.4	1.2	not interesting – interesting	Stimulation
2.2	1.2	1.1	unpredictable – predictable	Dependability
1.4	0.8	0.9	fast – slow	Efficiency
2.8	0.2	0.4	inventive – conventional	Novelty
1.4	3.0	1.7	obstructive – supportive	Dependability
2.4	2.8	1.7	good – bad	Attractiveness
1.4	1.8	1.3	complicated – easy	Perspicuity
2.4	0.5	0.7	unlikable – pleasing	Attractiveness
2.4	1.0	1.0	usual – leading edge	Novelty
2.9	0.4	0.6	unpleasant – pleasant	Attractiveness
2.9	0.1	0.3	secure – not secure	Dependability
2.0	0.8	0.9	motivating – demotivating	Stimulation
2.9	0.1	0.3	meets expectations – does not.	Dependability
0.4	1.8	1.3	inefficient – efficient	Efficiency
2.8	0.2	0.4	clear – confusing	Perspicuity
1.9	1.4	1.2	impractical – practical	Efficiency
2.8	0.4	0.7	organized – cluttered	Efficiency
2.6	0.5	0.7	attractive – unattractive	Attractiveness
2.2	2.9	1.7	friendly – unfriendly	Attractiveness
2.2	1.7	1.3	conservative – innovative	Novelty

Table 16: UEQ Scales - Mean and Variance

UEQ Scales	Mean	Std. Dev.
Attractiveness	2.426	0.13
Perspicuity	2.250	0.20
Efficiency	1.639	0.28
Dependability	2.361	0.47
Stimulation	2.194	0.26
Novelty	2.444	0.26